Sea Surface Wind Speed Retrieval From Textures in Synthetic Aperture Radar Imagery

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***Abstract*— Wind-induced oriented textures (WIOTs) are commonly used to retrieve sea surface wind directions from synthetic aperture radar (SAR) images. In this study, we found that WIOTs are also related to sea surface wind speeds (SSWSs). The entropy values in the gray-level cooccurrence matrices (GLCMs) for SAR images containing WIOTs will become steady with increasing distance between pairs of pixels. Furthermore, these steady values of entropy (SVEs) show a clear linear relationship with SSWSs. As a result, an SSWS retrieval model was developed based on this relationship. We used 2222/2223 Sentinel-1 SAR images (wind speed ranges from 5 to 20 m/s) to fit/validate the algorithm. The retrieved SSWSs were compared with the European Centre for Medium-Range Weather Forecast (ECMWF) SSWSs, Cross-Calibrated Multi-Platform (CCMP) SSWSs, and Tropical Atmosphere/Ocean (TAO) buoy measurements, and the root-mean-square differences (RMSDs) were 1.78, 1.70, and 1.78 m/s, respectively. The new model was also tested for SAR images acquired under hurricane conditions. The wind comparisons against stepped-frequency microwave radiometer (SFMR) measurements show an RMSD of 1.28 m/s. Our model's performance was also tested with the images at different spatial scales in the validation data set. Since the model is based on inherent image patterns, it still works well for SAR images without precise calibration.**

***Index Terms*— Entropy, gray-level cooccurrence matrix (GLCM), synthetic aperture radar (SAR), sea surface wind speed (SSWS), textures.**

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I. Introduction

WIND is the driving force that is directly responsible for atmospheric and oceanic motions. Sea surface wind plays an important role in many marine phenomena studies, for example, ocean circulations and waves [1]–[8]. Conventional wind measurements, made by ships and buoys, can provide sea surface wind data with only limited spatial coverage. In contrast, much wider coverage can be achieved through scatterometers [9], [10], radiometers [11], [12], and synthetic aperture radar (SAR) [13]–[15]. Owing to its high resolution and day–night imaging capability, SAR can capture surface roughness patterns modulated by many interesting atmospheric and oceanic phenomena, including tropical cyclones, atmospheric gravity waves, atmospheric vortex streets, internal waves, coastal upwellings, and so on [16]–[24].

Texture, an intrinsic property of any visual surface, has been extensively studied in numerous fields [25]–[29], among which considerable attention was paid to the extraction of texture features. The conventional gray-level cooccurrence matrix (GLCM) represents the joint probability distribution of the gray levels of pixel pairs (satisfying specific relative positions) within an image [30]. As a useful tool for texture analysis, GLCMs are usually used to derive statistical parameters that quantitatively describe different texture features and are widely used in image recognition, classification, and segmentation tasks.

Texture features often indicate information concerning some physical phenomena [31]. For example, sea surface wind streaks and marine atmospheric boundary layer rolls can produce oriented textures in SAR images, and these textures are generally aligned with sea surface wind directions [32]–[35]. Therefore, the sea surface wind direction can be retrieved from SAR imagery by wind-induced oriented textures (WIOTs), for which several methods have been developed, for example, the Fourier transform approach, local gradients method, and GLCM-based method [36]–[39]. However, the extraction of potential sea surface wind speed (SSWS) information hidden within WIOTs has not yet been investigated.

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The conventional retrieval of SSWSs from SAR images is based on geophysical model functions, which mainly formulate the dependence of the normalized radar cross section on the SSWS and the observation geometry (e.g., incidence angle and azimuth angle with respect to the wind direction) [40]–[46]. However, the context of WIOTs is not considered in this scheme.

Our study shows that the GLCM-derived entropy of SAR images containing WIOTs will become steady as the distance between a pair of pixels increases. Moreover, the steady value of entropy (SVE) is linearly correlated with the SSWS up to 20 m/s. Accordingly, an SSWS retrieval model is proposed based on this relationship and is tested at different spatial scales. The retrieval results of our model agree well with the SSWSs obtained from the European Centre for Medium-Range Weather Forecasts (ECMWF), Cross-Calibrated Multi-Platform (CCMP), and Tropical Atmosphere/Ocean (TAO) buoys. The new model also works well under hurricane conditions given the occurrence of WIOTs. Furthermore, our model's robustness against inaccurate radiometric calibration is tested, and the model still works well for SAR images without precise calibration.

The remainder of this article is organized as follows. In Section II, the data used in this article are introduced. In Section III, the procedure used to extract steady values from GLCMs is presented, and the linear relationship between the SVEs and ECMWF SSWSs is revealed. In Section IV, the above-mentioned linear relationship is validated. A new SSWS retrieval model based on this relationship is proposed and tested at different spatial scales. Our model is also tested under hurricane conditions and for inaccurate radiometric calibrations of SAR images. In Section V, discussions and summary are given.

II. Data

A. Sentinel-1 SAR Images

To investigate the relationship between the GLCM-derived parameters and SSWSs, we used the VV-polarized Sentinel-1 wave mode SAR imagery data set of wind streaks. This data set is provided by Institut français de recherche pour l'exploitation de la mer (IFREMER) and is publicly available at Sea Scientific Open Data Publication (SEANOE): *http://www.seanoe.org/data/00456/56796/*. Four criteria define the wind streaks in this data set: periodic linear features that dominate whole images; wavelengths between 0.8 and 5 km; homogeneous intensity modulation within a single scene; and the coexistence of ocean waves with periodic signatures [48]. These SAR images are collected in 20 km × 20 km scenes at two incidence angles of 23.8° (wave mode 1, WV1) and 36.8° (wave mode 2, WV2) and processed into Georeferenced Tagged Image File Format (GeoTIFF) with a resolution of 50 m by recalibration, downsampling, and normalization. To reduce the incidence angle effect, we recalibrated the normalized radar cross section (σ_0) by dividing the C-band model 5.N (CMOD5.N) simulated value of σ_0 (setting a constant SSWS of 10 m/s at 45° relative to the antenna look angle). The recalibrated σ_0 commonly ranges from 0 to 3. After performing the recalibration, the resolution of the images is downsampled to 50 m by an averaging operation to highlight larger features and simultaneously reduce speckle noise. Finally, to maintain all texture features and radiometric information, we normalized all images into 16-bit gray images by rescaling [0, 3] to [0, 65535]. Four sample images from the data set are shown in Fig. 1. The data set acquired in Sentinel-1 WV2 mode was randomly divided into two subsets. One subset is for finding the relationship between the GLCM-derived parameters and SSWSs. The other is for validation. These two data sets are named the fitting and validation data sets, respectively.

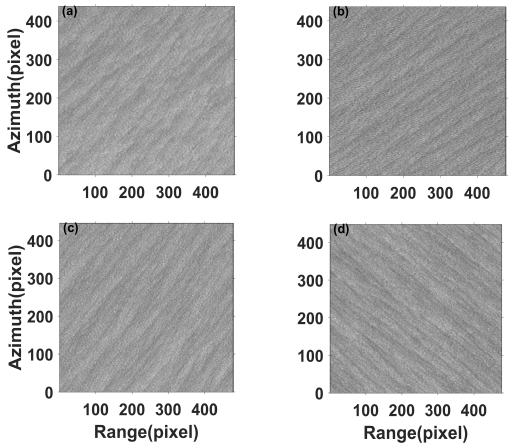

Fig. 1. Samples from the labeled Sentinel-1 wave mode wind streak data set. (a) Image at 20:36 UTC on January 1, 2016. (b) Image at 01:26 UTC on February 1, 2016. (c) Image at 14:36 UTC on August 2, 2016. (d) Image at 08:43 UTC on December 2, 2016.

B. ECMWF Reanalysis Data

The ECMWF Reanalysis (ERA)-Interim global atmospheric data set is produced with data assimilation using 12-hourly analysis cycles. In each cycle, available observations and prior information from a forecast model are combined to obtain the global atmosphere's evolutionary states and its underlying surface. Therefore, the data assimilation produces 6-hourly gridded products at different spatial resolutions and includes various kinds of global surface parameters, for example, u and v wind components at the height of 10 m on 0.125° grids. ERA-Interim covers a long period starting in 1979 and continues updating in near real-time. The width of wave mode is 20 km. The spatial resolution of ECMWF data is 12.5 km, and the temporal interval of ECMWF data is 6 h. Thus, each wave mode SAR image may cover one to four grids. In our study, we chose the nearest ECMWF grid to the center of the SAR image and temporally interpolated the ECMWF wind speed to the imaging time. Thus, only one collocated point can be obtained from one wave mode SAR image. The following describes the detailed procedure of collocating SAR images and ECMWF SSWSs. First, the nearest grid is located according to the center of an SAR image's latitude and longitude. Then, the wind components (u and v) at the imaging time are obtained by interpolating the ECMWF wind components between the nearest two times. Finally, the SSWSs are computed from the interpolated wind components.

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C. CCMP Wind Product

The CCMP wind product is provided by Remote Sensing Systems and has led to some interesting findings by Chelton, Hoffman, and others. Thus, we also used the SSWSs from the CCMP wind product for validation. The CCMP data set combines cross-calibrated satellite microwave winds with *in situ* observations to produce high-resolution gridded analysis data using a 4-D variational analysis method with the ERA-Interim model wind field as the background [47]. Each daily data file contains three arrays of dimensions 1440 (longitude) by 628 (from 78.375°S to 78.375°N) by 4 [at times of 0, 6, 12, and 18 Universal Time Coordinated (UTC)]. Two of the arrays comprise u and v wind components in meters per second at the height of 10 m. The absence of observation means that the wind vector for this grid was only obtained from the background field. Therefore, only the CCMP data with nonzero observation numbers are taken into consideration. It means that the CCMP products we used are close to actual wind speeds. The SAR/CCMP collocating procedure is the same as that used for the SAR/ECMWF data and there are 1290 collocated points.

D. Tropical Atmosphere Ocean (TAO) Buoy Measurements

The TAO system is an array of moored buoys in the tropical Pacific Ocean operated by the National Oceanic and Atmospheric Administration (NOAA) Pacific Marine Environment Laboratory. TAO buoys collect oceanographic and meteorological data for monitoring, forecast, and climate research [49]. The SSWSs included in these data need to be converted into equivalent neutral winds at 10 m for comparison. The SAR images in the validation data set are matched with the TAO buoys using a window of 0.2° in both longitude and latitude. The width of wave mode is small (20 km). Although we matched all Sentinel-1 wave mode SAR images provided by IFREMER with the buoy data, only four images are collocated with buoys.

E. SFMR Measurements

The stepped-frequency microwave radiometers (SFMRs) onboard the hurricane research aircraft operated by the NOAA and U.S. Air Force measure nadir microwave emissions at 6 C-band frequencies (4.55, 5.06, 5.64, 6.34, 6.96, and 7.22 GHz). The SFMR can provide along-track wind speed measurements at relatively high spatial (∼120 m) and temporal (1 s) resolutions and has been well-validated against both dropwindsonde and *in situ* instrument data [43], [44]. In this study, the SFMR measurements are collocated with four Sentinel-1A dual-polarized SAR images of Hurricanes Dorian and Michael. The time window for collocating the SFMR measurements is 30 min centered on the imaging time. Each SAR image is divided into 5 km × 5 km subimages to be matched with the SFMR tracks, and the collocated SFMR measurements of each matched subimage are averaged for comparison. Since the hurricane continues moving during the time window, the SFMR location is adjusted according to the velocity calculated by the hurricane's track from NOAA and the time interval between the measurements and imaging time.

III. Retrieval Model

A. GLCM and GLCM-Derived Parameters

An asymmetrical GLCM represents the joint probability distribution of gray levels at the positions of a pair of pixels satisfying specific relative positions within an image [30]. In conventional GLCM analysis, only the GLCMs of horizontal, vertical, and diagonal relative positions are computed; however, Zheng *et al.* [30] extended conventional GLCM analysis to arbitrary relative positions. To quantitatively analyze texture features, a set of GLCM-derived parameters are defined that are useful in the recognition, classification, and segmentation of an image [26]–[30]. The following definitions describe four commonly used parameters, namely, contrast, energy, entropy, and sum variance (SV):

$$\text{Contrast: } f_1(r,\theta) = \sum_{n=0}^{N-1} n^2 \left(\sum_{|i-j|=n} G(i,j;r,\theta) \right) \tag{1}$$

$$\text{Energy: } f_2(r,\theta) = \sum_i \sum_j [G(i,j;r,\theta)]^2 \tag{2}$$

$$\text{Entropy: } f_3(r,\theta) = -\sum_i \sum_j G(i,j;r,\theta) \log(G(i,j;r,\theta)) \tag{3}$$

$$\text{SV: } f_4(r,\theta) = \sum_i \sum_j (i-\mu)^2 G(i,j;r,\theta) \tag{4}$$

where $G(i, j; r, \theta)$ is the element of the GLCM with position $(i, j; r, \theta)$, N is the number of gray levels, and μ is the mean gray level of the image.

We converted all SAR images to 4-bit (0–15) gray images by a linear transformation to reduce the computational burden. Then, the GLCMs of each image are computed over distances and angles varying from 1 to 60 pixels and 0° to 359°, respectively. The distance step is 1 pixel, and the angle step is 1°. We calculated GLCM-related parameters using (1)–(4) and the texture directions using an existing method [30].

B. Steady Trend of GLCM-Derived Parameters (Energy, SV, and Entropy)

Four examples are provided in Fig. 2 to illustrate the steady trend of the GLCM-derived parameters, that is, energy (a), SV (b), and entropy (c). To determine the steady values of energy, SV, and entropy, we applied the following procedure. First, the parameter values in the direction perpendicular to the WIOT are arranged according to distance. Second, the mean and standard deviation (SD) of the series of seven consecutive parameter values are calculated, and the series is moved by one distance step until the SD is smaller than 1% of the mean. Third, the mean is considered a steady value. For clarity, Fig. 3 presents a flowchart of the steady value determination procedure for an SAR image.

C. Linear Relationship Between SVEs and SSWSs

Using the above procedure, the steady values of energy, SV, and entropy were estimated from the SAR images in the fitting data set. A scatter plot of these steady values

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TABLE I

CORRELATION COEFFICIENTS BETWEEN THE SVES AND THE REFERENCE DATA AND THE STATISTICS (CORRELATION COEFFICIENT, BIAS, AND RMSD) BETWEEN THE RETRIEVED SSWSS AND THE REFERENCE DATA

Dataset	Statistics	Reference Data	
		ECMWF	CCMP
Fitting (2222)	Correlation Coefficient	0.7131	0.7395
Validation (2223)	Correlation Coefficient	0.7054	0.7601
	Bias (m/s)	-0.08	-0.51
	RMSD (m/s)	1.78	1.70

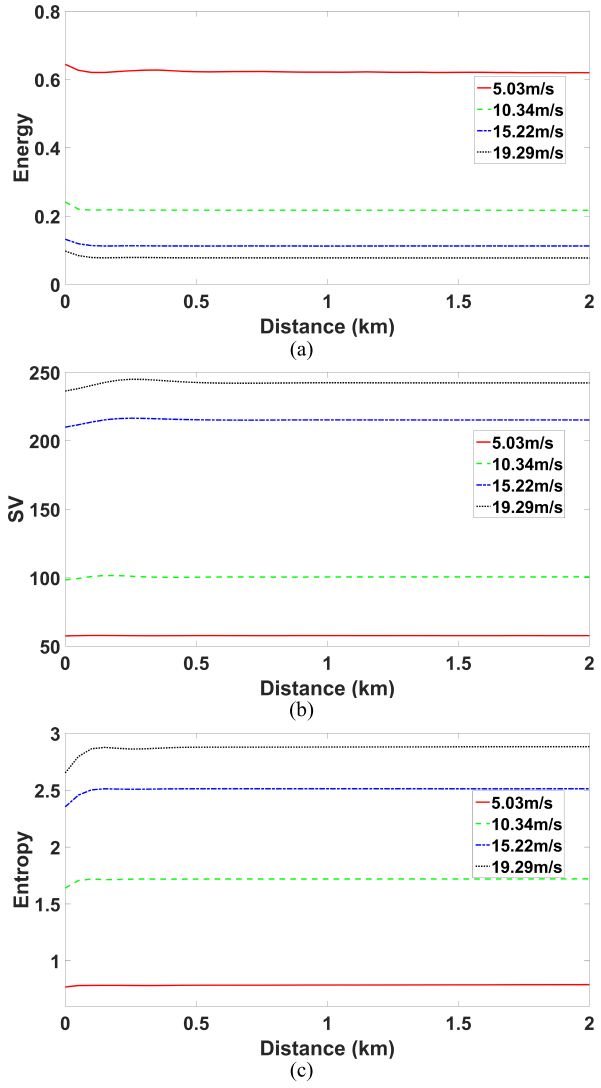

Fig. 2. Four examples of the GLCM-derived parameters that vary with distance at different SSWSs. (a) Energy. (b) SV. (c) Entropy.

versus the collocated ECMWF SSWSs is shown in Fig. 4. To quantitatively evaluate the linear relationship between the SVEs and ECMWF SSWSs, the correlation coefficients are calculated and shown in Fig. 4. The SVEs show a much higher linear correlation with the ECMWF SSWSs than the other steady values, and the fitting formula is

$$y = 4.4676x + 1.7286 \tag{5}$$

where x and y represent the SVEs and SSWSs, respectively. Entropy measures the disorder of an image. When the image is not texturally uniform, many GLCM elements have very small values, which implies that entropy is very large [50].

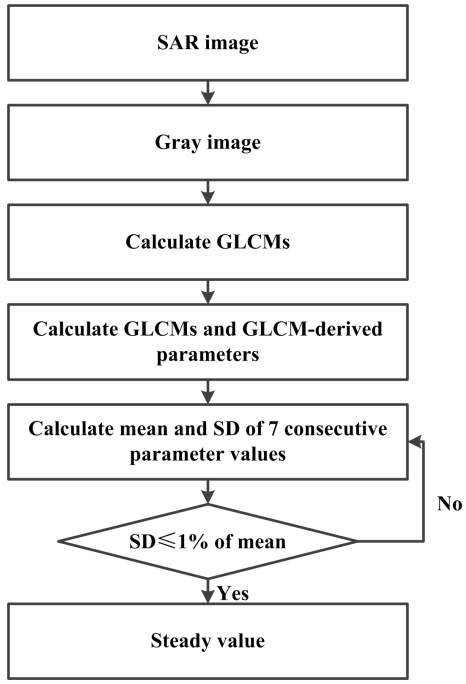

Fig. 3. Flowchart of extracting steady value from an SAR image.

The disorder of an SAR image with WIOTs will become large with the increase in the SSWS.

IV. VALIDATION AND TEST

A. Accuracy Validation

The SAR images in the validation data set were used to validate the developed relationship in (5). The SVEs of these images were calculated and used to retrieve the SSWSs by applying (5). The retrieved SSWSs were compared with the collocated ECMWF SSWSs and the collocated CCMP SSWSs. The results in Fig. 5 show that the retrieved SSWSs are in good agreement with both the ECMWF SSWSs and the CCMP SSWSs. The root-mean-square differences (RMSDs) between the retrieved SSWSs and the ECMWF SSWSs (1.78 m/s) and CCMP SSWSs (1.70 m/s) are shown in Table I.

We compared the retrieved SSWSs of the validation data set with the collocated TAO buoy measurements, and the results are shown in Table II. The retrieved SSWSs are in good agreement with the collocated TAO buoy measurements with an RMSD of 1.78 m/s. The bias is 1.42 m/s, and the 95% confidential interval is from –0.51 to 3.35 m/s.

We also investigated the performance of our model at different spatial scales. For this purpose, smaller images were cropped from the original SAR images in the validation data set at scales of 15 km ×15 km, 10 km × 10 km, and

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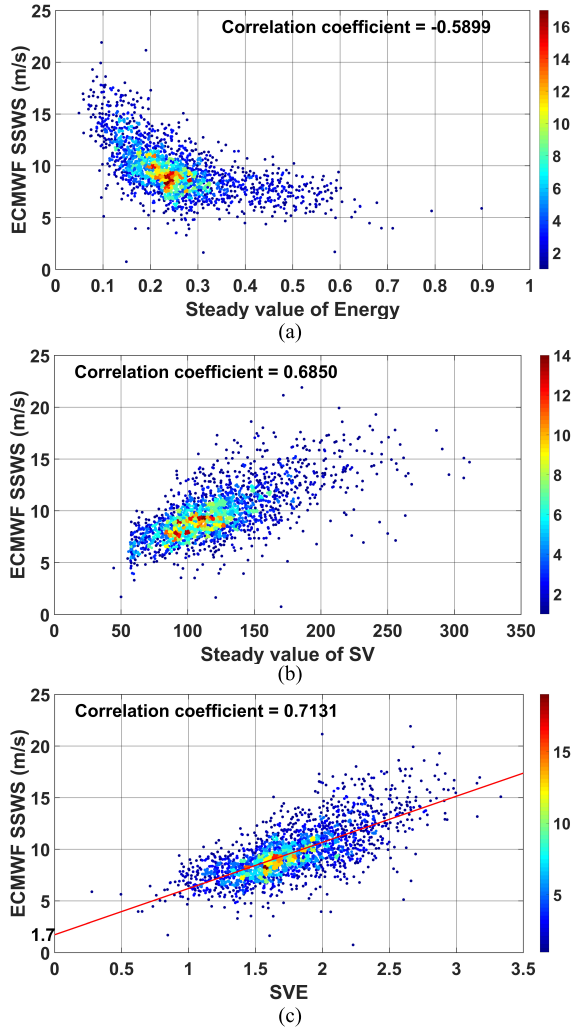

Fig. 4. Relationship between three types of steady values and ECMWF SSWSs. (a) Energy. (b) SV. (c) SVE. The colorbar ranges from blue (low counts) to red (high counts), and there are 2222 counts in total.

5 km × 5 km, but the image centers remained unchanged. Then, the SVEs of the new images were calculated, and the SSWSs were retrieved by applying the new model. Finally, the RMSDs between the retrieved SSWSs at each scale and the ECMWF SSWSs were computed and compared. The results are shown in Fig. 6. The RMSD shows a slight increase with a decrease in the spatial scale, but the changes are not significant. This fact demonstrates that the new model works well at different spatial scales.

B. Tests Under Hurricane Conditions

Considering that WIOTs may occur during hurricanes, our model was tested in four dual-polarized [VV and vertical–horizontal (VH)] Sentinel-1A SAR images of Hurricanes Michael and Dorian. Hurricane Michael made landfall as an unprecedented Category 5 Hurricane in the Florida Panhandle region with maximum sustained wind speeds of 140 knots. The storm caused catastrophic damage from wind and storm surge, particularly in the Panama City Beach to Mexico Beach to Cape San Blas areas. The widespread damage spread well inland as Hurricane Michael remained at hurricane strength into southwest Georgia. Hurricane Dorian was the strongest

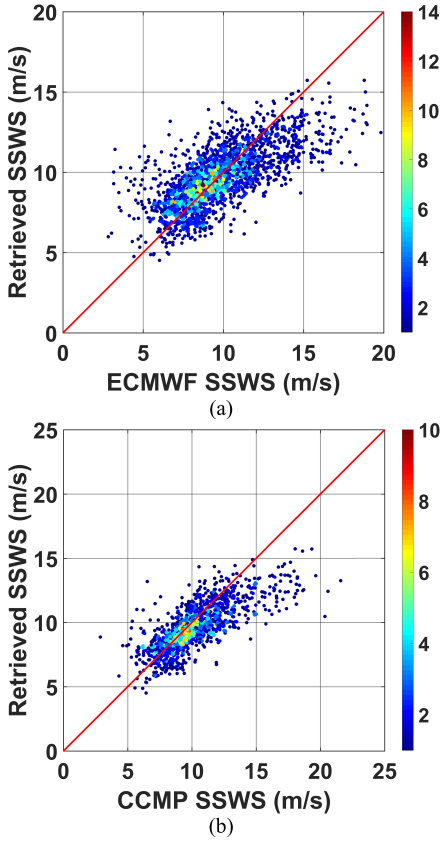

Fig. 5. Comparisons between the retrieved SSWSs by the new model and the collocated reference data: (a) ECMWF, (b) CCMP, and there are 1290 counts in total.

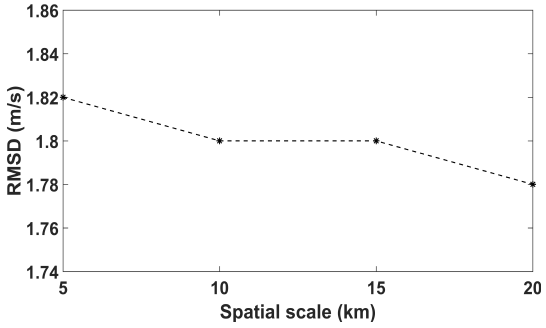

Fig. 6. RMSDs between the retrieved SSWSs by the new model and the collocated ECMWF SSWSs at different spatial scales.

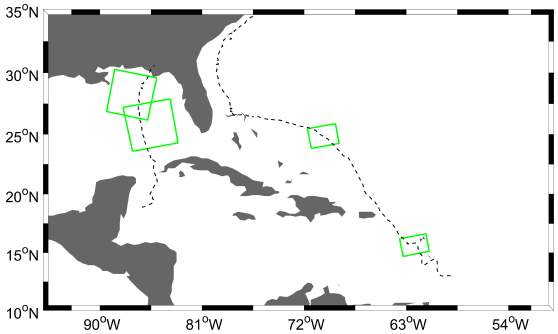

Fig. 7. Locations of four Sentinel-1A dual-polarized SAR images during hurricanes. Hurricanes' tracks are denoted by black dash lines.

and most destructive storm of the 2019 hurricane season. The northern Bahamas, the Abaco Islands, and Grand Bahama

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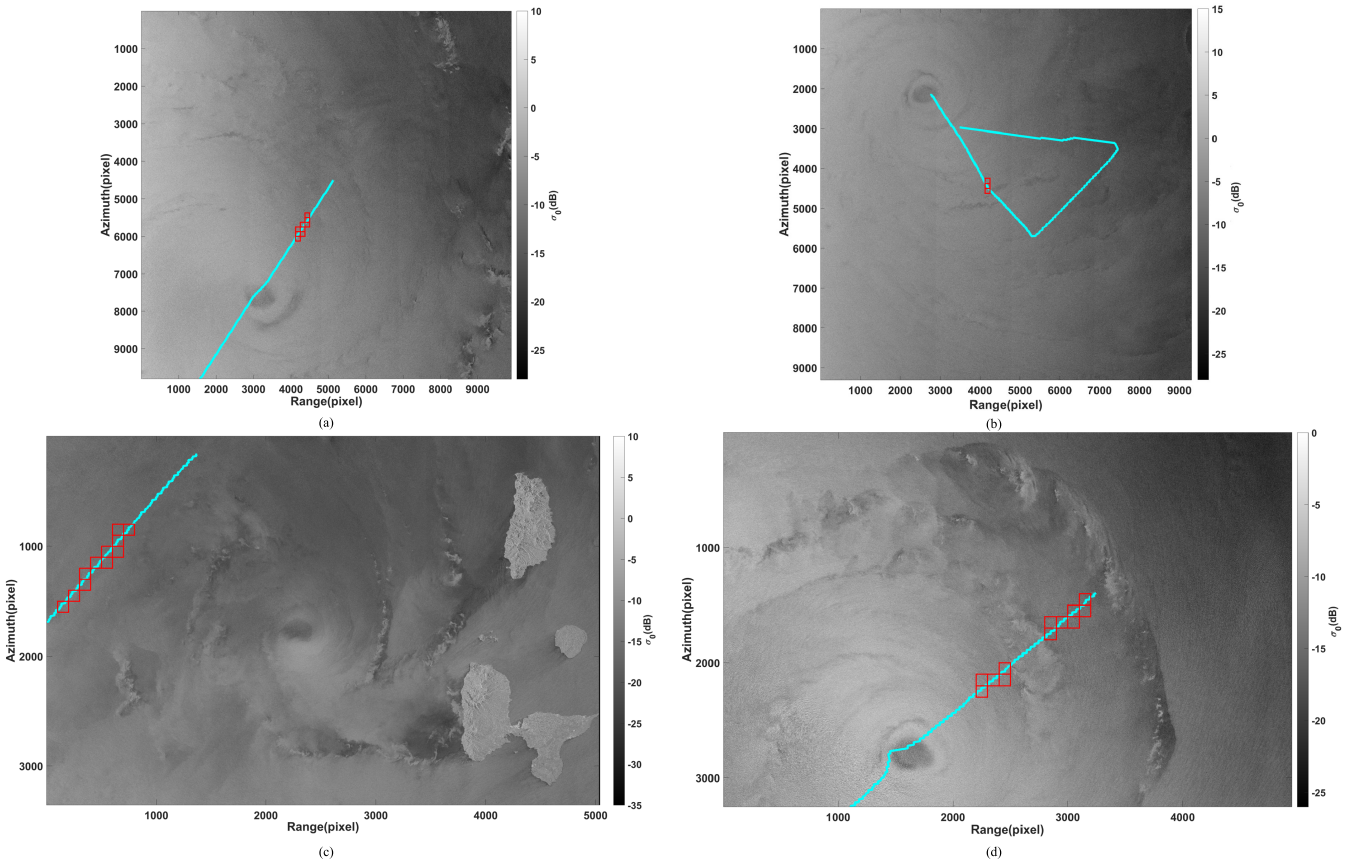

Fig. 8. Sentinel-1A VV-polarized SAR images. (a) Hurricane Michael (22:19 UTC, October 9, 2018). (b) Hurricane Michael (11:50 UTC, October 10, 2018). (c) Hurricane Dorian (22:19 UTC, August 27, 2019). (d) Hurricane Dorian (22:46 UTC, August 30, 2019). The resolution of these SAR images is 50 m. The cyan lines represent the SFMR tracks and the red boxes represent the collocated subimages (5 km × 5 km) with clear WIOTs.

TABLE II

INFORMATION OF THE BUOYS AND COLLOCATED SAR IMAGES IN THE VALIDATION DATA SET AND A COMPARISON BETWEEN THE TAO BUOY-MEASURED SSWSS AND RETRIEVED SSWSS

Buoy			SAR image			SSWS (m/s)		
Station	Lat.	Lon.	Time (UTC)	Lat.	Lon.	Buoy	Retrieved	Diff.
2S125W	2.07°S	124.94°W	06/07/2016, 02:37:44	1.99°S	124.75°W	5.90	8.79	2.89
8S155W	8.26°S	155.05°W	15/09/2016, 15:59:13	8.37°S	155.08°W	10.10	10.04	-0.06
2S125W	2.06°S	124.94°W	16/09/2016, 02:37:47	1.99°S	124.75°W	5.06	6.28	1.22
2S125W	2.07°S	124.93°W	22/10/2016, 02:37:47	1.99°S	124.75°W	6.35	7.98	1.63

Island were devastated by Dorian's category five winds estimated to have reached over 180 mph with a storm surge greater than 18 feet. The locations of the four images are shown in Fig. 7 with the green boxes. The images are shown in Fig. 8 with the SFMR tracks denoted by cyan lines. The red boxes represent the 5 km × 5 km subimages, which display clear WIOTs and match up well with the SFMR tracks.

The recalibration of these subimages is in the same way as the wave mode for consistency. The SSWSs in the red boxes were calculated by the existing wind retrieval algorithms CMOD7, C2POD, C3PO, and our model and compared them with the SFMR measurements. The cyan diamonds, blue pluses, magenta crosses, and red circles denote the results of CMOD7, C3PO, C2POD, and our model in Fig. 9, respectively, and Table III gives the statistics. The SSWSs calculated by our model are relatively consistent with the SFMR SSWSs with a bias of −0.48 m/s and an RMSD of 1.28 m/s. Most of the CMOD7 SSWSs are underestimated compared with the SFMR SSWSs with a bias of –4.97 m/s and an RMSD of 5.73 m/s. The C2POD and C3PO SSWSs agree well with the SFMR SSWSs for the relatively high SSWSs (>15 m/s) but are overestimated for the relatively low SSWSs (<15 m/s). The biases of the C2POD and C3PO SSWSs are 1.36 and 1.40 m/s, respectively, and their RMSDs are 3.35 and 2.94 m/s, respectively.

C. Test of Robustness Against Inaccurate Radiometric Calibration

Typically, the SAR radiometric calibration has a linear-mapping form as

$$\sigma_0 = (1+\alpha)p \cdot I + (q+\beta) \tag{6}$$

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TABLE III
STATISTICS BETWEEN THE SSWSS CALCULATED BY DIFFERENT MODELS AND THE COLLOCATED SFMR SSWSS

Model	Statistics (10-27m/s)		
	Correlation Coefficient	Bias (m/s)	RMSD (m/s)
CMOD7	0.8171	-4.97	5.73
C3PO	0.8823	1.40	2.94
C2POD	0.8129	1.36	3.35
New model	0.9710	-0.48	1.28

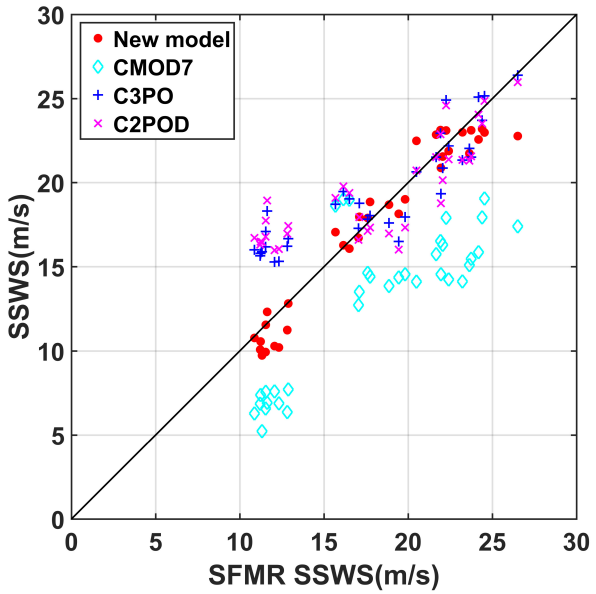

Fig. 9. Comparisons of SSWSs calculated by CMOD7, C2POD, C3PO, and the new model with the SFMR SSWSs. The red circles, cyan diamonds, blue pluses, and magenta crosses denote SSWSs calculated by the new model, CMOD7, C3PO, and C2POD, respectively. There are 35 points for each model.

where I represents the intensity of the image, p and q are the additive factor and the multiplicative factor, respectively, and α and β are the changes in p and q, respectively. A change in the additive factor leads only to a small change in the entropy because of the overflow or underflow in the gray level and rounding operation. Therefore, the SVE derived from the entropy will have a small change as well. The following test was performed to test the robustness of (5) against the corruption of an inaccurate additive factor in a radiometric calibration. First, a constant β (6000) was added to the SAR images in the fitting and validation data sets. Then, the SVEs of the SAR images were determined by the two procedures described in Section III-B and III-C. For the fitting data set, a scatter plot of the SVEs versus the collocated ECMWF SSWSs is shown in Fig. 10(a). The red solid line in Fig. 10(a) is the original fitted line determined in Section III-C, and the red dashed line represents the new line fitted to the scatters. These two lines have only a slight difference. For the validation data set, the SVEs were further used to retrieve the SSWSs using (5). The retrieved SSWSs are still in good agreement with the ECMWF SSWSs as shown in Fig. 11(a).

The robustness of (5) was also tested against the corruption of an inaccurate multiplicative factor in the radiometric calibration. First, the SAR images' gray levels were decreased by a factor of 0.8 ($\alpha = -0.2$) for both the fitting and the

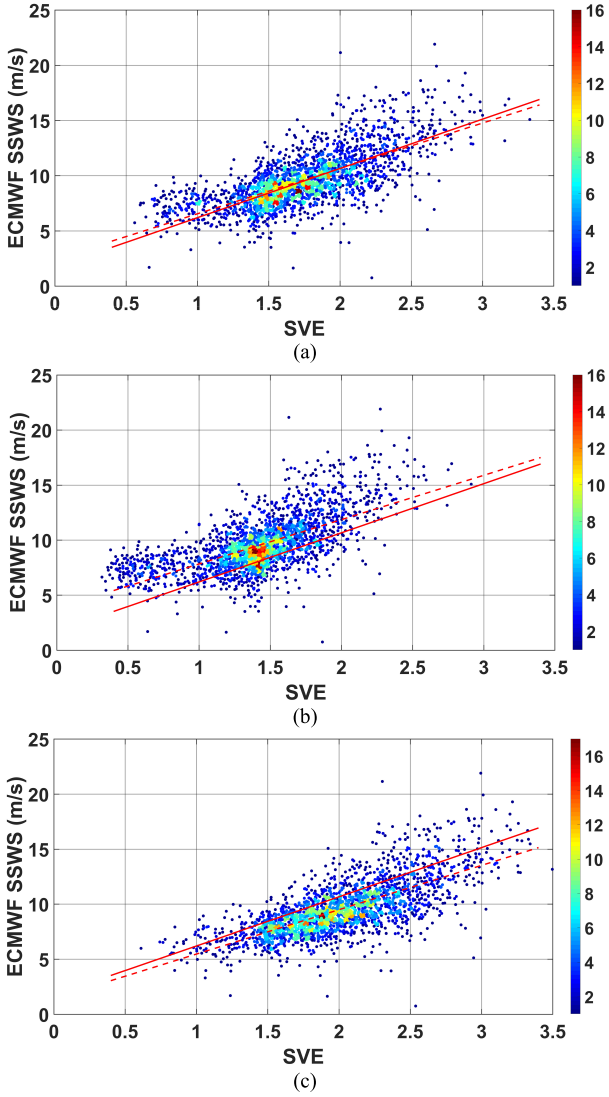

Fig. 10. Affection of the inaccurate calibrations for the fitting data set. (a) Additive factor change ($\beta = 6000$). (b) Multiplicative factor change ($\alpha = -0.2$). (c) Multiplicative factor change ($\alpha = 0.2$).

validation data sets. Then, the SVEs of the SAR images were determined, and the SSWSs were further retrieved for the validation data set. For the fitting data set, the scatter plot of the SVEs versus the collocated ECMWF SSWSs is shown in Fig. 10(b), in which there is an apparent discrepancy between the solid line (the original fitted line determined in Section III-C) and the dashed line (the new line fitted to the scatters). For the validation data set, Fig. 11(b) shows that the retrieved SSWSs are slightly lower overall than the ECMWF SSWSs. The correlation coefficient computed between the

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TABLE IV

CORRELATION COEFFICIENTS BETWEEN THE SVES AND THE REFERENCE DATA FOR THE FITTING DATA SET WITH AN INACCURATE CALIBRATION (FOR ADDITIVE CORRUPTION, $\alpha = 0$, $\beta = 6000$; FOR MULTIPLICATIVE CORRUPTION 1, $\alpha = -0.2$, $\beta = 0$; FOR MULTIPLICATIVE CORRUPTION 2, $\alpha = 0.2$, $\beta = 0$)

Dataset	Calibration Corruption	Reference Data	
		ECMWF	CCMP
Fitting	Additive	0.6996	0.7366
	Multiplicative 1	0.6521	0.6897
	Multiplicative 2	0.7194	0.7516

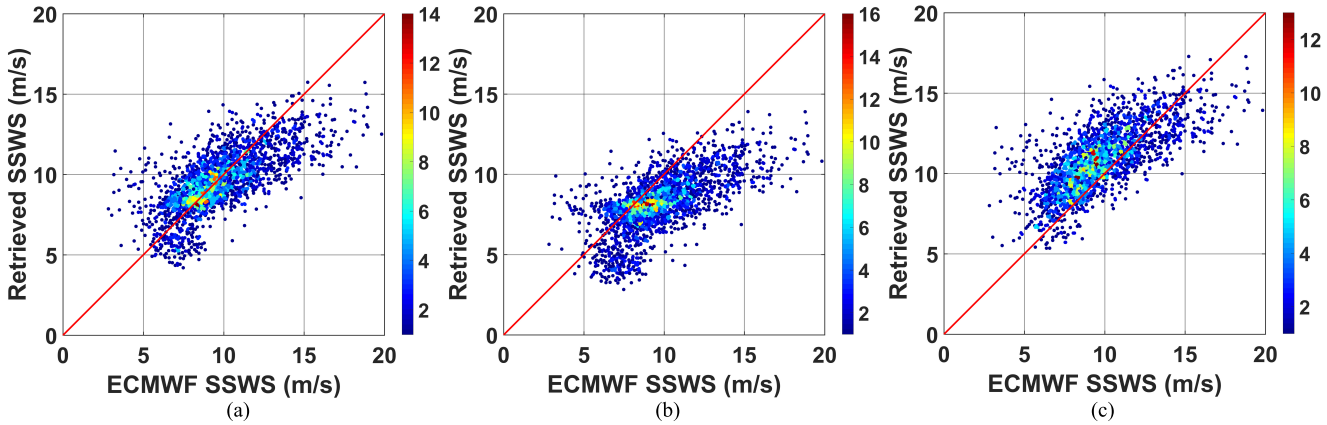

Fig. 11. Comparisons between the retrieved SSWSs of validation data set and ECMWF SSWSs while the calibration is inaccurate. (a) Additive factor change ($\beta = 6000$). (b) Multiplicative factor change ($\alpha = -0.2$). (c) Multiplicative factor change ($\alpha = 0.2$).

SVEs and ECMWF SSWSs for the fitting data set is shown in Table IV.

The SAR images' gray levels were also increased by a factor of 1.2 ($\alpha = 0.2$) for both the fitting and the validation data sets for further testing. The same procedure was carried out, and the results are shown in Figs. 10(c) and 11(c) and Table IV. Similarly, there is an apparent discrepancy between the solid line and the dashed line in Fig. 10(c). The retrieved SSWSs of the validation data set are slightly larger than the ECMWF SSWSs in Fig. 11(c). The correlation coefficient, bias, and RMSD between the retrieved SSWSs and ECMWF SSWSs were also computed for the validation data set, and the results are shown in Table V. The CCMP SSWSs were further taken as a reference to test the robustness of (5) against inaccurate radiometric calibrations. Similar results were obtained, as shown in Fig. 12 and Tables IV and V.

V. DISCUSSION AND SUMMARY

It should be noted that the SAR images' homogeneity has been improved by recalibration, which can reduce the incidence angle effect along the range direction. The performance of our model under hurricane conditions is reasonably well when compared with other models' results. However, the full hurricane wind field cannot be obtained by our model since the WIOTs are impossible to occur all over the SAR image. Meanwhile, the robust automatic extraction of subimages that contain wind-induced textures and distinguishing wind-induced textures from others in SAR images are still a challenge. Thus, the extraction was done visually in our research. The multiplicative factor's effect is more evident than the additive factor in the calibration procedure, especially for the case that $\alpha < 0$. The results in Section IV-C suggest that texture features can be well-preserved when $\alpha > 0$ and will be partially lost when $\alpha < 0$.

The WIOTs in SAR imagery will be affected by the polarization [51], [52]. However, in a common procedure of wind speed retrieval, the horizontal–horizontal (HH)-polarized SAR images are converted to the VV-polarized SAR images by polarization ratio models, first. The error induced by the polarization ratio can be equivalent to the error caused by the incorrect multiplicative factor in the radiometric calibration, which has been tested in Section IV-C of the article.

Attention should always be paid to wind direction information for the sake of ascertaining the WIOTs in SAR imagery. However, our study discovered that WIOTs also contain SSWS information. There is a clear linear relationship between the GLCM-derived SVEs of WIOTs and SSWSs, and this relationship can be used to retrieve SSWSs from SAR imagery. We built a linear SSWS retrieval model from the fitting data set based on this relationship, and the model was applied to the validation data set. The retrieved SSWSs were compared with the ECMWF SSWSs, CCMP SSWSs, and TAO buoy measurements, and the RMSDs were 1.78, 1.70, and 1.78 m/s, respectively. Moreover, the model was shown to work well at different spatial scales. Under hurricane conditions, the SSWSs retrieved by the new model were relatively consistent with the collocated SFMR SSWSs with an RMSD of 1.28 m/s. The model also worked well in robustness tests against inaccurate additive and multiplicative factors in

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TABLE V

STATISTICS (CORRELATION COEFFICIENT, BIAS, AND RMSD) BETWEEN THE RETRIEVED SSWSS AND THE REFERENCE DATA FOR THE VALIDATION DATA SET WITH AN INACCURATE CALIBRATION (FOR ADDITIVE CORRUPTION, $\alpha = 0$, $\beta = 6000$; FOR MULTIPLICATIVE CORRUPTION 1, $\alpha = -0.2$, $\beta = 0$; FOR MULTIPLICATIVE CORRUPTION 2, $\alpha = 0.2$, $\beta = 0$)

Dataset	Calibration Corruption	Statistic	Reference Data	
			ECMWF	CCMP
Validation	Additive	Correlation Coefficient	0.6997	0.7543
		Bias (m/s)	-0.12	-0.60
		RMSD (m/s)	1.80	1.74
	Multiplicative 1	Correlation Coefficient	0.6582	0.7117
		Bias (m/s)	-1.49	-1.97
		RMSD (m/s)	2.40	2.63
	Multiplicative 2	Correlation Coefficient	0.7147	0.7671
		Bias (m/s)	1.11	0.66
		RMSD (m/s)	2.08	1.73

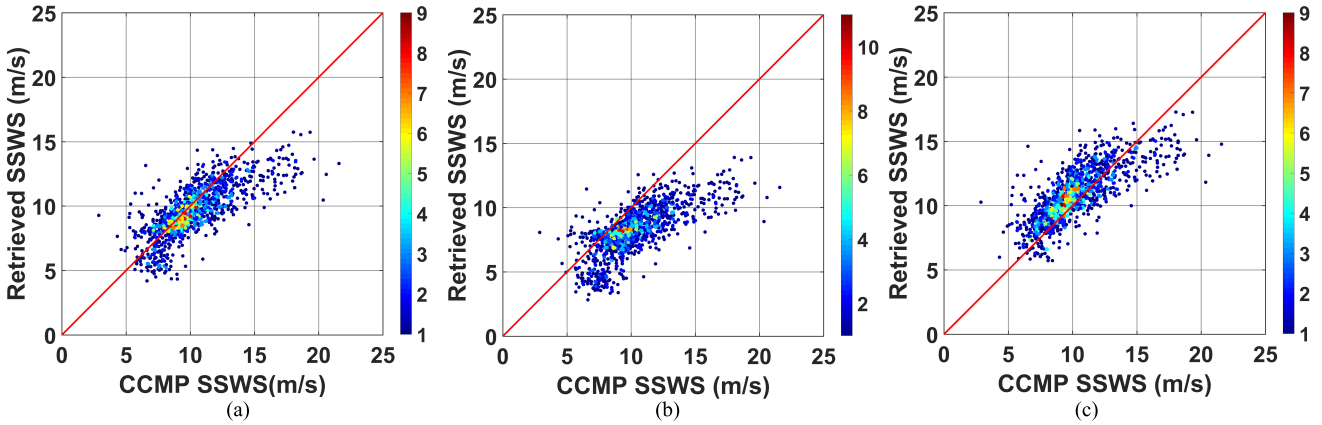

Fig. 12. Comparisons between the retrieved SSWSs of validation data set and CCMP SSWSs while the calibration is inaccurate. (a) Additive factor change ($\beta = 6000$). (b) Multiplicative factor change ($\alpha = -0.2$). (c) Multiplicative factor change ($\alpha = 0.2$).

the radiometric calibration of SAR imagery. By the way, the wind range of Sentinel-1 wave mode SAR images is from 5 to 20 m/s, and the wind range of hurricane cases is from 10 to 27 m/s.

The SSWS intercept of the fitted line in the model is 1.73 m/s. Linear fitting of the validation data set achieved a similar SSWS intercept of 1.75 m/s. Considering that SVEs are nonnegative, the above findings suggest that wind streaks are unlikely to emerge when the SSWS is less than 1.74 m/s (the average of 1.72 and 1.75 m/s).

To the best of our knowledge, this is the first study to reveal a relationship between the SSWSs and WIOTs in SAR imagery by a GLCM-derived parameter, and this strongly linear relationship can be used to retrieve SSWSs from SAR imagery.

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REFERENCES

[1] J. Jiang, "Estimating the energy flux from the wind to ocean inertial motions: The sensitivity to surface wind fields," *Geophys. Res. Lett.*, vol. 32, no. 15, pp. 291–310, 2005.

[2] P. P. Sullivan and J. C. McWilliams, "Dynamics of winds and currents coupled to surface waves," *Annu. Rev. Fluid Mech.*, vol. 42, no. 1, pp. 19–42, Jan. 2010.

[3] S. K. Satheesh and D. Lubin, "Short wave versus long wave radiative forcing by indian ocean aerosols: Role of sea-surface winds," *Geophys. Res. Lett.*, vol. 30, no. 13, pp. 56–67, Jul. 2003, doi: 10.1029/2003gl017499.

[4] L. Cavaleri *et al.*, "Wave modelling—The state of the art," *Prog. Oceanogr.*, vol. 75, no. 4, pp. 603–674, Dec. 2007.

[5] H. Hu, "Oceanic thermal and biological responses to santa ana winds," *Geophys. Res. Lett.*, vol. 30, no. 11, pp. 389–401, 2003.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 08:08:22 UTC from IEEE Xplore. Restrictions apply.

[6] J. M. Von Ahn, J. M. Sienkiewicz, and P. S. Chang, "Operational impact of QuikSCAT winds at the NOAA ocean prediction center," *Weather Forecasting*, vol. 21, no. 4, pp. 523–539, Aug. 2006.

[7] Q. Xu, X. Li, Y. Wei, Z. Tang, Y. Cheng, and W. G. Pichel, "Satellite observations and modeling of oil spill trajectories in the Bohai Sea," *Mar. Pollut. Bull.*, vol. 71, nos. 1–2, pp. 107–116, Jun. 2013.

[8] Y. Cheng *et al.*, "Monitoring of oil spill trajectories with COSMO-SkyMed X-band SAR images and model simulation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 7, pp. 2895–2901, Jul. 2014.

[9] X. Yang, X. Li, Q. Zheng, X. Gu, W. G. Pichel, and Z. Li, "Comparison of ocean-surface winds retrieved from QuikSCAT scatterometer and Radarsat-1 SAR in offshore waters of the U.S. West Coast," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 1, pp. 163–167, Jan. 2011.

[10] R. Atlas *et al.*, "The effects of marine winds from scatterometer data on weather analysis and forecasting," *Bull. Amer. Meteorological Soc.*, vol. 82, no. 9, pp. 1965–1990, Sep. 2001.

[11] T. Meissner, L. Ricciardulli, and F. J. Wentz, "Capability of the SMAP mission to measure ocean surface winds in storms," *Bull. Amer. Meteorological Soc.*, vol. 98, no. 8, pp. 1660–1677, Sep. 2017.

[12] G. Zheng, J. Yang, L. Ren, W. Zhou, and L. Huang, "The preliminary cross-calibration of the HY-2A calibration microwave radiometer with the Jason-1/2 microwave radiometers," *Int. J. Remote Sens.*, vol. 35, nos. 11–12, pp. 4515–4531, Jun. 2014.

[13] K. S. Friedman and X. Li, "Monitoring hurricanes over the ocean with wide swath SAR," *Johns Hopkins APL Tech. Dig.*, vol. 21, no. 1, pp. 80–85, Jan. 2000.

[14] M. Bruunchristiansen, W. Koch, J. Horstmann, C. Bayhasager, and M. Nielsen, "Wind resource assessment from C-band SAR," *Remote Sens. Environ.*, vol. 105, no. 1, pp. 68–81, Nov. 2006.

[15] K. S. Friedman, T. D. Sikora, W. G. Pichel, P. Clemente-Colón, and G. Hufford, "Using spaceborne synthetic aperture radar to improve marine surface analyses," *Weather Forecasting*, vol. 16, no. 2, pp. 270–276, Apr. 2001.

[16] X. Li *et al.*, "Observation of hurricane-generated ocean swell refraction at the gulf stream north wall with the RADARSAT-1 synthetic aperture radar," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 10, pp. 2131–2142, Jan. 2002.

[17] G. Zheng, J. Yang, A. K. Liu, X. Li, W. G. Pichel, and S. He, "Comparison of typhoon centers from SAR and IR images and those from best track data sets," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 2, pp. 1000–1012, Feb. 2016.

[18] R. Romeiser *et al.*, "On the remote sensing of oceanic and atmospheric convection in the greenland sea by synthetic aperture radar," *J. Geophys. Res., Oceans*, vol. 109, no. C3, pp. 1–14, Mar. 2004, doi: 10.1029/2003jc001975.

[19] X. Li, W. Zheng, W. G. Pichel, C.-Z. Zou, and P. Clemente-Colón, "Coastal katabatic winds imaged by SAR," *Geophys. Res. Lett.*, vol. 34, no. 3, pp. 300–315, Feb. 2007.

[20] Y. Zhao, X.-M. Li, and J. Sha, "Sea surface wind streaks in spaceborne synthetic aperture radar imagery," *J. Geophys. Res., Oceans*, vol. 121, no. 9, pp. 6731–6741, Sep. 2016.

[21] X. Li, W. Zheng, X. Yang, J. A. Zhang, W. G. Pichel, and Z. Li, "Coexistence of atmospheric gravity waves and boundary layer rolls observed by SAR," *J. Atmos. Sci.*, vol. 70, no. 11, pp. 3448–3459, Nov. 2013.

[22] X. Li, "Synthetic aperture radar observation of the sea surface imprints of upstream atmospheric solitons generated by flow impeded by an island," *J. Geophys. Res.*, vol. 109, no. C2, pp. 235–250, 2004.

[23] R. Chang, R. Zhu, M. Badger, C. Hasager, X. Xing, and Y. Jiang, "Offshore wind resources assessment from multiple satellite data and WRF modeling over south China sea," *Remote Sens.*, vol. 7, no. 1, pp. 467–487, Jan. 2015.

[24] S. Dinardo *et al.*, "Coastal SAR and PLRM altimetry in German bight and West Baltic sea," *Adv. Space Res.*, vol. 62, no. 6, pp. 1371–1404, Sep. 2018.

[25] F. T. Ulaby, F. Kouyate, B. Brisco, and T. H. L. Williams, "Textural information in SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. GE-24, no. 2, pp. 235–245, Mar. 1986.

[26] R. N. Sutton and E. L. Hall, "Texture measures for automatic classification of pulmonary disease," *IEEE Trans. Comput.*, vol. C-21, no. 7, pp. 667–676, Jul. 1972.

[27] R. Parekh, "Using texture analysis for medical diagnosis," *IEEE MultiMediaMag.*, vol. 19, no. 2, pp. 28–37, Feb. 2012.

[28] R. Krishnan K and S. Radhakrishnan, "Focal and diffused liver disease classification from ultrasound images based on isocontour segmentation," *IET Image Process.*, vol. 9, no. 4, pp. 261–270, Apr. 2015.

[29] H. Pan, P. Gao, H. Zhou, R. Ma, J. Yang, and X. Zhang, "Roughness analysis of sea surface from visible images by texture," *IEEE Access*, vol. 8, pp. 46448–46458, Mar. 2020.

[30] G. Zheng *et al.*, "Development of a gray-level co-occurrence matrix-based texture orientation estimation method and its application in sea surface wind direction retrieval from SAR imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5244–5260, Sep. 2018.

[31] S. Ufermann and R. Romeiser, "Numerical study on signatures of atmospheric convective cells in radar images of the ocean," *J. Geophys. Res., Oceans*, vol. 104, no. C11, pp. 25707–25719, Nov. 1999.

[32] D. Etling and R. A. Brown, "Roll vortices in the planetary boundary layer: A review," *Boundary-Layer Meteorol.*, vol. 65, no. 3, pp. 215–248, Aug. 1993.

[33] P. D. Mourad and B. A. Walter, "Viewing a cold air outbreak using satellite-based synthetic aperture radar and advanced very high resolution radiometer imagery," *J. Geophys. Res., Oceans*, vol. 101, no. C7, pp. 16391–16400, Jul. 1996.

[34] C. Wang *et al.*, "Characteristics of marine atmospheric boundary layer roll vortices from Sentinel-1 SAR wave mode," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Yokohama, Japan, Jul. 2019, pp. 7908–7911.

[35] R. C. Foster, F. Vianey, P. Drobinski, and P. Carlotti, "Near-surface coherent structures and the vertical momentum flux in a large-eddy simulation of the neutrally-stratified boundary layer," *Boundary-Layer Meteorol.*, vol. 120, no. 2, pp. 229–255, Aug. 2006.

[36] C. Wackerman, R. Shuchman, and F. Fetterer, "Estimation of wind speed and wind direction from ERS-1 imagery," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Pasadena, CA, USA, USA, Aug. 1994, vol. 2, pp. 1222–1224.

[37] J.-S. Yang, W.-G. Huang, C.-B. Zhou, B. Fu, A.-Q. Shi, and D.-L. Li, "Coastal ocean surface wind retrieval from SAR imagery," *J. Remote Sens.*, vol. 5, no. 1, pp. 16–20, 2001.

[38] W. Koch, "Directional analysis of SAR images aiming at wind direction," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 4, pp. 702–710, Apr. 2004.

[39] L. Zhou *et al.*, "An improved local gradient method for sea surface wind direction retrieval from SAR imagery," *Remote Sens.*, vol. 9, no. 7, p. 671, Jun. 2017.

[40] H. Hersbach, A. Stoffelen, and S. de Haan, "An improved C-band scatterometer ocean geophysical model function: CMOD5," *J. Geophys. Res.*, vol. 112, no. C3, pp. 1–18, 2007, doi: 10.1029/2006jc003743.

[41] H. Hersbach, "Comparison of C-Band scatterometer CMOD5.N equivalent neutral winds with ECMWF," *J. Atmos. Ocean. Technol.*, vol. 27, no. 4, pp. 721–736, Apr. 2010.

[42] B. Zhang, W. Perrie, and Y. He, "Wind speed retrieval from RADARSAT-2 quad-polarization images using a new polarization ratio model," *J. Geophys. Res.*, vol. 116, no. C8, pp. 1–13, 2011, doi: 10.1029/2010jc006522.

[43] G. Zhang, X. Li, W. Perrie, P. A. Hwang, B. Zhang, and X. Yang, "A hurricane wind speed retrieval model for C-Band RADARSAT-2 cross-polarization ScanSAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 8, pp. 4766–4774, Aug. 2017.

[44] B. Zhang, W. Perrie, J. A. Zhang, E. W. Uhlhorn, and Y. He, "High-resolution hurricane vector winds from C-band dual-polarization SAR observations," *J. Atmos. Ocean. Technol.*, vol. 31, no. 2, pp. 272–286, Feb. 2014.

[45] L. Ren *et al.*, "Preliminary analysis of chinese GF-3 SAR quad-polarization measurements to extract winds in each polarization," *Remote Sens.*, vol. 9, no. 12, p. 1215, Nov. 2017.

[46] L. Ren *et al.*, "Assessments of ocean wind retrieval schemes used for chinese Gaofen-3 synthetic aperture radar co-polarized data," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 7075–7085, Sep. 2019.

[47] R. Atlas *et al.*, "A cross-calibrated, multiplatform ocean surface wind velocity product for meteorological and oceanographic applications," *Bull. Amer. Meteorological Soc.*, vol. 92, no. 2, pp. 157–174, Feb. 2011.

[48] C. Wang *et al.*, "A labelled ocean SAR imagery dataset of ten geophysical phenomena from Sentinel-1 wave mode," *Geosci. Data J.*, vol. 6, no. 2, pp. 105–115, Nov. 2019.

[49] L. LeBlanc *et al.*, "Results of the national data buoy center (NDBC) commissioning process on the tropical atmosphere ocean (TAO) technology refreshed system," in *Proc. OCEANS MTS/IEEE SEATTLE*, Seattle, WA, USA, Sep. 2010, pp. 1–6.

[50] A. Baraldi and F. Parmiggiani, "An investigation of the textural characteristics associated with gray level cooccurrence matrix statistical parameters," *IEEE Trans. Geosci. Remote Sens.*, vol. 33, no. 2, pp. 293–304, Mar. 1995.

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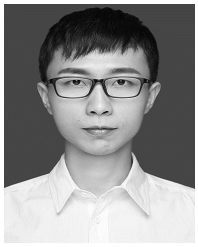
[51] G. Zhang, W. Perrie, B. Zhang, S. Khurshid, and K. Warner, "Semi-empirical ocean surface model for compact-polarimetry mode SAR of RADARSAT constellation mission," *Remote Sens. Environ.*, vol. 217, pp. 52–60, Nov. 2018.
[52] G. Zhang *et al.*, "C-band right-circular polarization ocean wind retrieval," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 9, pp. 1398–1401, Sep. 2019.



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